FOOD PROCESSING, COLD CHAINS AND VALUE ADDITION

FARM-TO-VALUE RAIL → COLD-CHAIN CONTINUITY → CAPACITY ACCOUNTING FORK → PMKSY UMBRELLA → SCHEME-STAGE LADDER → OPERATION GREENS BOUNDARY

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

★ PRIMARY CORE FLOW · ◆ SUBORDINATE EXTRA · ● APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

00

STAGE 00 FARM-TO-VALUE RAIL

FARM-TO-VALUE RAIL · FARM OUTPUT · AGGREGATION + GRADING · AMBIENT LOGISTICS + PROCESSING · PACKAGING + MARKET · DISTRIBUTED VALUE · MUST REMEMBER · FOOD PROCESSING ADDS VALUE THROUGH SORTING, GRADING, STORAGE

FARM OUTPUT → AGGREGATION + GRADING → COLD / AMBIENT LOGISTICS + PROCESSING → PACKAGING + MARKET → DISTRIBUTED VALUE → MUST REMEMBER: Food processing adds value through sorting, grading, storage

CORE CLAIM
- FARM OUTPUT
- AGGREGATION + GRADING

MECHANISM
- COLD / AMBIENT LOGISTICS + PROCESSING
- PACKAGING + MARKET → DISTRIBUTED VALUE

CONSEQUENCE / LIMIT
- MUST REMEMBER: Food processing adds value through sorting, grading, storage,...
- FARM OUTPUT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Food processing adds value through sorting, grading, storage..

01

STAGE 01 COLD-CHAIN CONTINUITY

COLD-CHAIN CONTINUITY · PACK HOUSE · COLD STORAGE · REEFER → PROCESSOR / RETAIL

PACK HOUSE → COLD STORAGE → REEFER → PROCESSOR / RETAIL

CORE CLAIM
- PACK HOUSE
- COLD STORAGE

MECHANISM
- REEFER → PROCESSOR / RETAIL
- PACK HOUSE

CONSEQUENCE / LIMIT
- COLD STORAGE
- REEFER → PROCESSOR / RETAIL

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRE-COOLING; COLD STORAGE; REEFER → PROCESSOR / RETAIL.

02

STAGE 02 CAPACITY ACCOUNTING FORK

CAPACITY ACCOUNTING FORK · INSTALLED CAPACITY · OPERATIONAL CAPACITY · AVAILABLE STOCK · PERIOD FLOW · VALUE ADDED · OUTPUT VALUE MINUS INPUTS

INSTALLED CAPACITY → stock

OPERATIONAL CAPACITY → available stock → THROUGHPUT → period flow → VALUE ADDED → output value minus inputs

CORE CLAIM
- INSTALLED CAPACITY → stock
- OPERATIONAL CAPACITY → available stock

MECHANISM
- THROUGHPUT → period flow
- VALUE ADDED → output value minus inputs

CONSEQUENCE / LIMIT
- INSTALLED CAPACITY → stock
- OPERATIONAL CAPACITY → available stock

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: THROUGHPUT → period flow; VALUE ADDED → output value minus inputs.

03

STAGE 03 PMKSY UMBRELLA

PMKSY UMBRELLA · CLUSTER INFRASTRUCTURE · COLD CHAIN + VALUE ADDITION · PROCESSING + QUALITY SUPPORT · GUIDELINE AND COMPONENT VINTAGE

COMMON / CLUSTER INFRASTRUCTURE

COLD CHAIN + VALUE ADDITION → PROCESSING + QUALITY SUPPORT → STATUS → guideline and component vintage

CORE CLAIM
- COMMON / CLUSTER INFRASTRUCTURE
- COLD CHAIN + VALUE ADDITION

MECHANISM
- PROCESSING + QUALITY SUPPORT
- STATUS → guideline and component vintage

CONSEQUENCE / LIMIT
- COMMON / CLUSTER INFRASTRUCTURE
- COLD CHAIN + VALUE ADDITION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PROCESSING + QUALITY SUPPORT; STATUS → guideline and component vintage.

04

STAGE 04 SCHEME-STAGE LADDER

SCHEME-STAGE LADDER · OPERATION + CAPACITY USE · APPROVAL / SANCTION · INVESTMENT / CONSTRUCTION

APPROVAL / SANCTION

INVESTMENT / CONSTRUCTION → OPERATION + CAPACITY USE → SALES / DISBURSAL / OUTCOME

CORE CLAIM
- APPROVAL / SANCTION
- INVESTMENT / CONSTRUCTION

MECHANISM
- OPERATION + CAPACITY USE
- SALES / DISBURSAL / OUTCOME

CONSEQUENCE / LIMIT
- APPROVAL / SANCTION
- INVESTMENT / CONSTRUCTION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OPERATION + CAPACITY USE; SALES / DISBURSAL / OUTCOME.

05

STAGE 05 OPERATION GREENS BOUNDARY

OPERATION GREENS BOUNDARY · PERISHABLE PRICE STRESS · SPECIFIED COMMODITY · STORAGE + PROCESSING LINK · NOT UNIVERSAL FARM-PRICE INSURANCE · CLOSE DISTINCTION · PROCESSING IS NOT ONLY MANUFACTURING, COLD STORAGE IS NOT

PERISHABLE PRICE STRESS → SPECIFIED COMMODITY / WINDOW → STORAGE + PROCESSING LINK → NOT universal farm-price insurance → CLOSE DISTINCTION: Processing is not only manufacturing, cold storage is not an end-to

CORE CLAIM
- PERISHABLE PRICE STRESS
- SPECIFIED COMMODITY / WINDOW

MECHANISM
- STORAGE + PROCESSING LINK
- NOT universal farm-price insurance

CONSEQUENCE / LIMIT
- CLOSE DISTINCTION: Processing is not only manufacturing, cold storage is not an end-to-...
- PERISHABLE PRICE STRESS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Processing is not only manufacturing, cold storage is not an end-to-.

06

STAGE 06 FOOD PARK CLUSTER

FOOD PARK CLUSTER · COLLECTION CENTRE · PRIMARY PROCESSING · COMMON CENTRAL FACILITIES · PROCESSOR + MARKET LINK

COLLECTION CENTRE

PRIMARY PROCESSING → COMMON CENTRAL FACILITIES → PROCESSOR + MARKET LINK

CORE CLAIM
- COLLECTION CENTRE
- PRIMARY PROCESSING

MECHANISM
- COMMON CENTRAL FACILITIES
- PROCESSOR + MARKET LINK

CONSEQUENCE / LIMIT
- COLLECTION CENTRE
- PRIMARY PROCESSING

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRIMARY PROCESSING; COMMON CENTRAL FACILITIES; PROCESSOR + MARKET LINK.

07

STAGE 07 REGULATORY-EXPORT FORK

REGULATORY-EXPORT FORK · FOOD-SAFETY REGULATION · EXPORT DEVELOPMENT · MARKET ACCESS · SPS + TESTING · COMPLIANCE GATE · CHAIN EVIDENCE

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
FSSAI → food-safety regulation	SPS + TESTING → compliance gate	FSSAI → food-safety regulation
APEDA → export development / market access	TRACEABILITY → chain evidence	APEDA → export development / market access

CORE CLAIM
- FSSAI → food-safety regulation
- APEDA → export development / market access

MECHANISM
- SPS + TESTING → compliance gate
- TRACEABILITY → chain evidence

CONSEQUENCE / LIMIT
- FSSAI → food-safety regulation
- APEDA → export development / market access

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SPS + TESTING → compliance gate; TRACEABILITY → chain evidence.

08

STAGE 08 VALUE-CAPTURE BOARD

VALUE-CAPTURE BOARD · RAW MATERIAL + AGGREGATION · TRANSFORMATION + QUALITY RISK · ACCESS + SHELF · CONTRACT + COMPETITION · MARGIN SPLIT

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
FARMER / FPO → raw material + aggregation	LOGISTICS / RETAIL → access + shelf	FARMER / FPO → raw material + aggregation
PROCESSOR → transformation + quality risk	CONTRACT + COMPETITION → margin split	PROCESSOR → transformation + quality risk

CORE CLAIM
- FARMER / FPO → raw material + aggregation
- PROCESSOR → transformation + quality risk

MECHANISM
- LOGISTICS / RETAIL → access + shelf
- CONTRACT + COMPETITION → margin split

CONSEQUENCE / LIMIT
- FARMER / FPO → raw material + aggregation
- PROCESSOR → transformation + quality risk

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FARMER / FPO → raw material + aggregation.

09

STAGE 09 EMPLOYMENT CHAIN

EMPLOYMENT CHAIN · FARM-GATE SORTING · COLD LOGISTICS + MAINTENANCE · PROCESSING + PACKAGING + TESTING · EXPORT; TEST FORMALITY + SEASONALITY

FARM-GATE SORTING

COLD LOGISTICS + MAINTENANCE → PROCESSING + PACKAGING + TESTING → RETAIL / EXPORT; test formality + seasonality

CORE CLAIM
- FARM-GATE SORTING
- COLD LOGISTICS + MAINTENANCE

MECHANISM
- PROCESSING + PACKAGING + TESTING
- RETAIL / EXPORT; test formality + seasonality

CONSEQUENCE / LIMIT
- FARM-GATE SORTING
- COLD LOGISTICS + MAINTENANCE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PROCESSING + PACKAGING + TESTING; RETAIL / EXPORT; test formality + seasonality.

10

STAGE 10 SMALL-PROCESSOR CONSTRAINT WHEEL

SMALL-PROCESSOR CONSTRAINT WHEEL · WORKING CAPITAL · POWER + TECHNOLOGY · TESTING + FORMALISATION · MARKET + BRAND + DISTRIBUTION

WORKING CAPITAL → POWER + TECHNOLOGY → TESTING + FORMALISATION → MARKET + BRAND + DISTRIBUTION

CORE CLAIM
- WORKING CAPITAL
- POWER + TECHNOLOGY

MECHANISM
- TESTING + FORMALISATION
- MARKET + BRAND + DISTRIBUTION

CONSEQUENCE / LIMIT
- WORKING CAPITAL
- POWER + TECHNOLOGY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TESTING + FORMALISATION; MARKET + BRAND + DISTRIBUTION.

11

STAGE 11 PROCESSING ANSWER SPINE

PROCESSING ANSWER SPINE · DEFINE CHAIN AND ACCOUNTING UNITS · MAP MISSING NODE AND SCHEME STAGE · TEST STANDARDS, JOBS AND BARGAINING · CONCLUDE WITH THROUGHPUT + FAIR VALUE CAPTURE · EVIDENCE LIMIT · USE COMMODITY-SPECIFIC CHAIN

DEFINE chain and accounting units → MAP missing node and scheme stage → TEST standards, jobs and bargaining → CONCLUDE with throughput + fair value capture → EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: Use commodity-specific chain

CORE CLAIM
- DEFINE chain and accounting units
- MAP missing node and scheme stage

MECHANISM
- TEST standards, jobs and bargaining
- CONCLUDE with throughput + fair value capture

CONSEQUENCE / LIMIT
- EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: Use commodity-specific chain,...
- DEFINE chain and accounting units

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: Use commodity-specific chain..

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · VIABILITY DEPENDS ON THROUGHPUT, SEASONALITY, CAPACITY UTILISATION, ENERGY COST · CLUSTER APPROACHES CAN CONNECT FARMS, COMMON TESTING, PACKAGING, LOGISTICS · CONTRACT DESIGN ALLOCATES QUALITY, QUANTITY, PRICE AND FORCE-MAJEURE RISK · GEOGRAPHICAL INDICATIONS, BRANDING AND TRACEABILITY CAN MOVE PRODUCERS FROM · FOOD SAFETY REGULATION ADDRESSES INFORMATION ASYMMETRY BECAUSE CONSUMERS CANNOT · BY-PRODUCTS CAN SUPPORT A CIRCULAR BIOECONOMY THROUGH FEED, ENERGY · MINISTRY OF FOOD PROCESSING INDUSTRIES

OPTIONAL SOURCE DEPTH
- CAUTION: Viability depends on throughput, seasonality, capacity utilisation, energy cost and
- CAUTION: Cluster approaches can connect farms, common testing, packaging, logistics, skills and
- CAUTION: Contract design allocates quality, quantity, price and force-majeure risk between
- CAUTION: Geographical indications, branding and traceability can move producers from commodity

ADVANCED DEBATE
- CAUTION: Food safety regulation addresses information asymmetry because consumers cannot easily
- CAUTION: By-products can support a circular bioeconomy through feed, energy, compost and
- Ministry of Food Processing Industries: frames sector programmes and infrastructure
- FSSAI: sets and enforces food-safety standards within its mandate.

LEGACY / NUANCE
- APEDA and commodity or export bodies: support standards, traceability and external
- FPOs, processors and logistics providers: coordinate procurement, throughput and
- CAUTION: A cold store without pre-cooling and reefer transport may merely shift the point at
- CAUTION: A small processor may have viable demand but remain constrained by testing cost,

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels